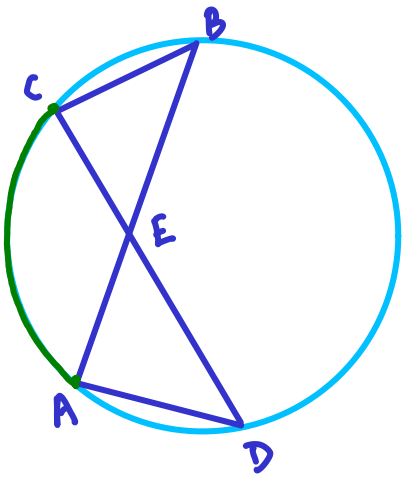
Circles

Segment Lengths in Circles



- Relation between $\triangle CBE$ and $\triangle ADE$?

  _____

- What proportions does the relation give?

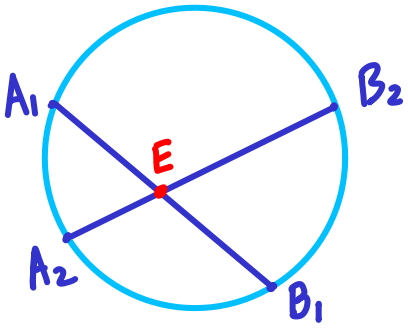
  $\frac{EC}{EA} =$ _____ or $\frac{EC}{EB} =$ _____

- What equation do the proportions give?

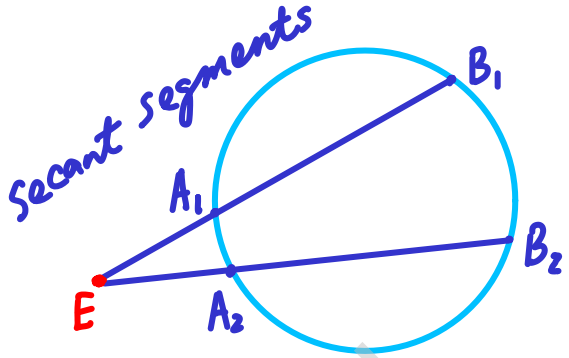
  $EC \cdot ED =$ _____

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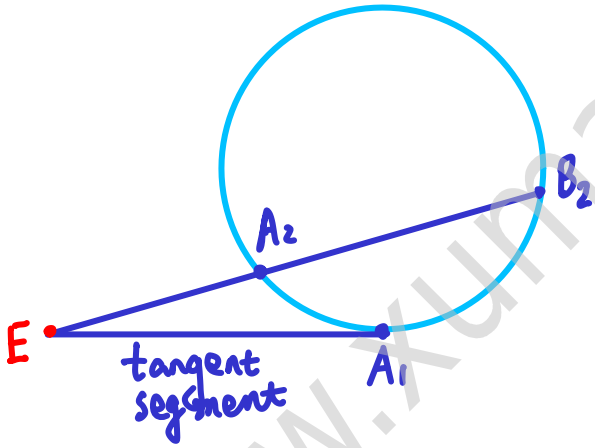
The Power of a Point Theorem !



I. $EA_1 \cdot EB_1 = EA_2 \cdot EB_2$



II. $EA_1 \cdot EB_1 = EA_2 \cdot EB_2$



III. $(EA_1)^2 = EA_2 \cdot EB_2$

As $EA_1 \cdot EB_1$ with $B_1 = A_1$

Warning !

Should be $EA_1 \cdot EB_1$

NOT $EA_1 \cdot A_1B_1$

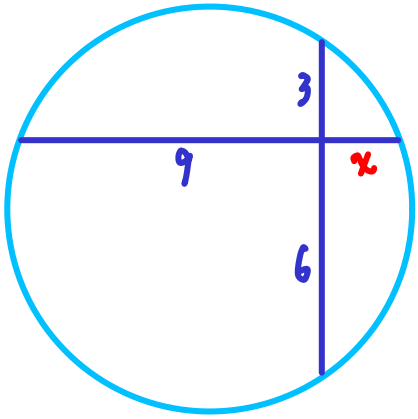
Note:
- A line through a point E intersects a circle at two points A and B. The product $EA \cdot EB$ is the same for all such lines.
- The product $EA \cdot EB$ is called the power of point E.

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Using the Power of a Point

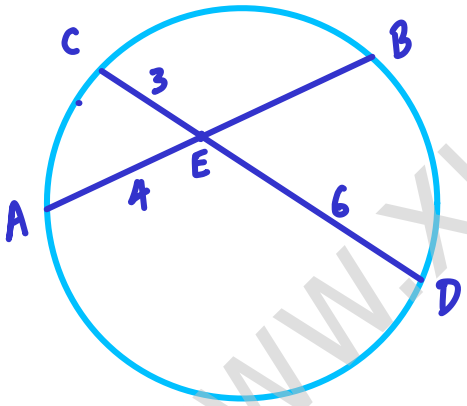
Practice

1.



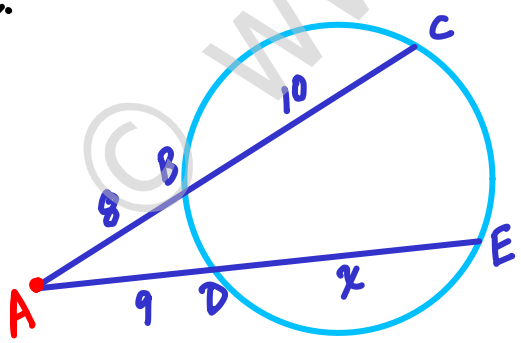
$x =$ _____

2.



$AB =$ _____

E.g.



$x = ?$

[Soln.:]

$AB \cdot AC = AD \cdot AE$

! NOT $AB \cdot BC$! NOT $AD \cdot DE$

Note: All segments from the point A

where $AC = 8 + 10 = 18$

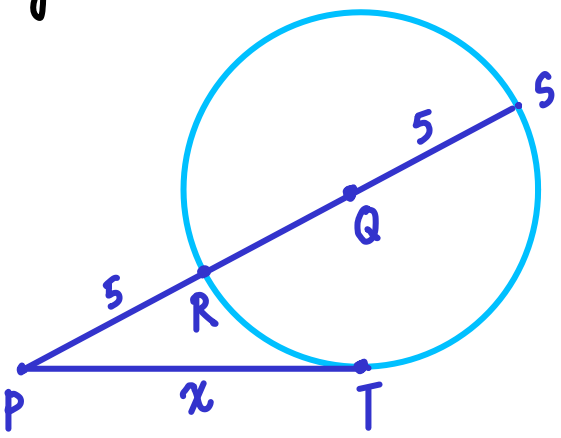
$AE = 9 + x$

So: $8 \cdot 18 = 9 \cdot (9 + x)$

$9 + x = \frac{8 \cdot 18}{9} = 16$, $x = 7$

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Eg.



$\odot Q$: center Q, radius = 5

$\overline{PT}$: tangent to $\odot Q$

Find x.

[Soln.:] $PT^2 = PR \cdot PS$

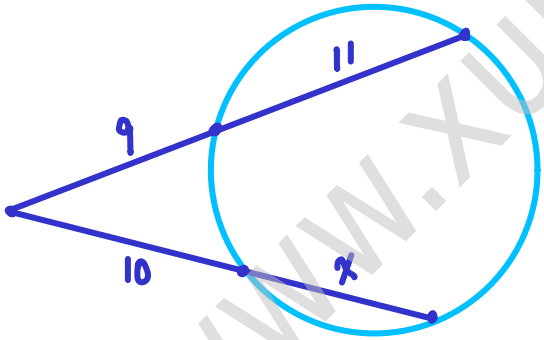
$PS = 5 + 10 = 15$

So $x^2 = 5 \times 15 = 5 \cdot (5 \times 3) = 5^2 \cdot 3$

$x = \sqrt{5^2 \cdot 3} = 5\sqrt{3}$

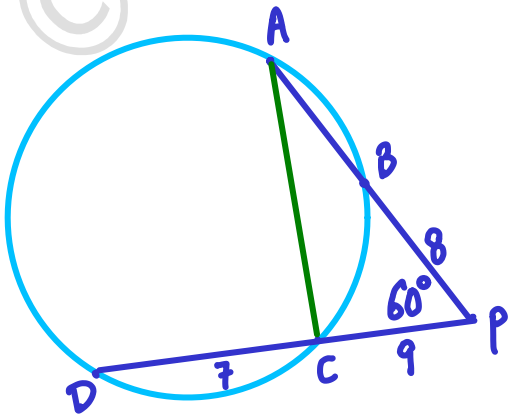
Practice

3.



$x =$ _____

*4.

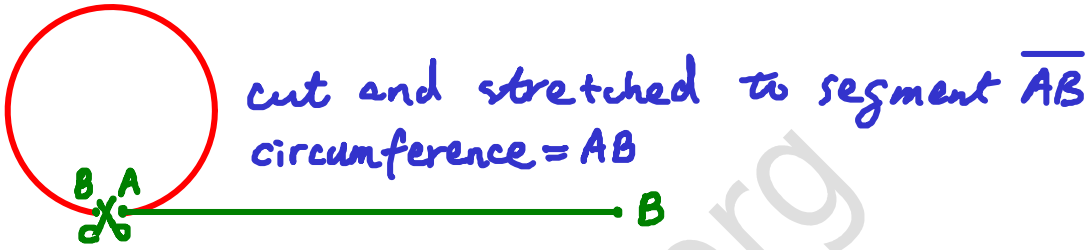


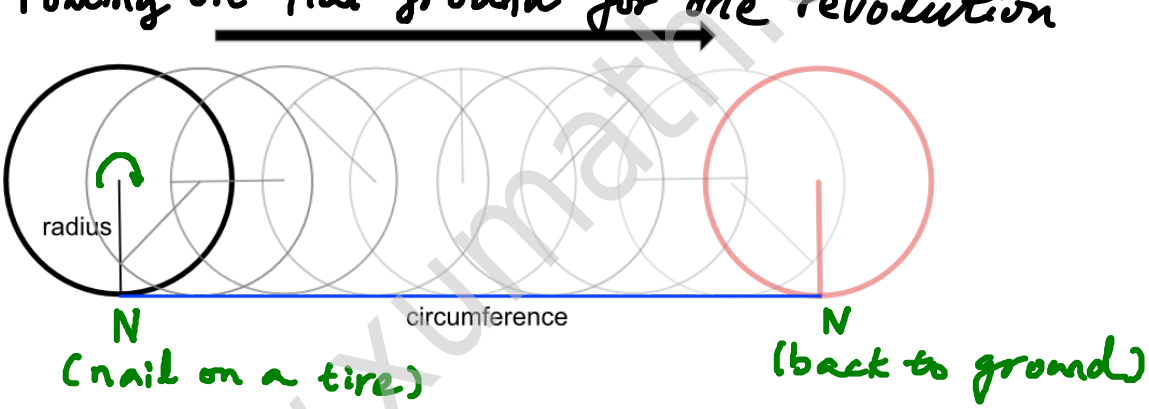
$\angle ACP =$ _____

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Circumference of a circle

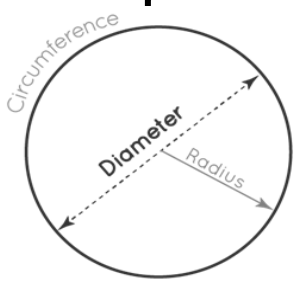
* The perimeter of a circle, which is the distance around the circle, is called the circumference of the circle.





Note: When a circle rolls on a stright line, the circle travels the distance of its circumference if a contact point between the circle and the line is back to be a contact point again for the first time.

* π : pai



$\frac{\text{Circumference}}{\text{Diameter}} = \pi = 3.14159\ldots$

$$\frac{\text{circumference}}{\text{diameter}} = \pi \approx 3.14$$

- independent of the circle size
- the constant π is irrational (as $\sqrt{2}$, not a quotient of two integers)

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Can a circle with an integer radius have an integer circumference? (yes or no) _____

* Summary:

Let radius = r, diameter = d, circumference = l

Then $\boxed{\begin{aligned} d &= 2r \\ l &= \pi d = 2\pi r \end{aligned}}$

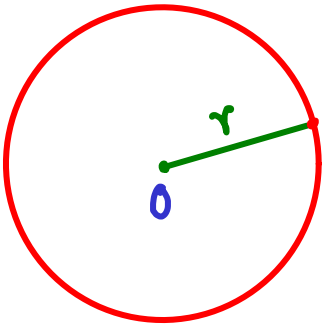
Practice

5. What is the radius of a circle that has circumference 54π centimeters? _____ cm

6. The tires of a car finish 10 revolutions (full rotations) in 1 second. If the radius of the tires is 16 inches, how many feet does the car travel on a flat road in 1 minute? $\underline{\quad\quad}\cdot\pi$ [ft].

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Area of a circle



* The area of a circle with radius r is

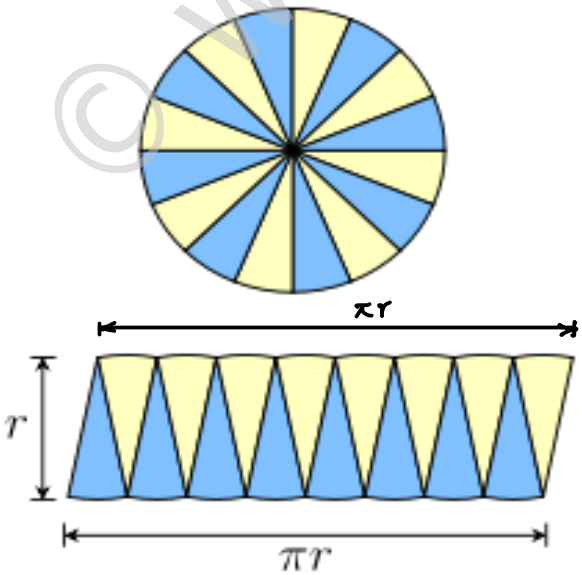
$$\boxed{A = \pi r^2}$$

Note:
- The area of a circle with diameter d is $\boxed{A = \frac{1}{4}\pi d^2}$

  as $\pi r^2 = \pi\left(\frac{d}{2}\right)^2 = \pi\frac{d^2}{2^2} = \frac{1}{4}\pi d^2$
- If the radius changes from r to $k \cdot r$, then the area changes from A to $k^2 \cdot A$

  as $\underbrace{\pi(kr)^2}_{\text{area with } kr} = \pi k^2 r^2 = k^2(\underbrace{\pi r^2}_{\text{area with } r})$

* Why $A = \pi r^2$?



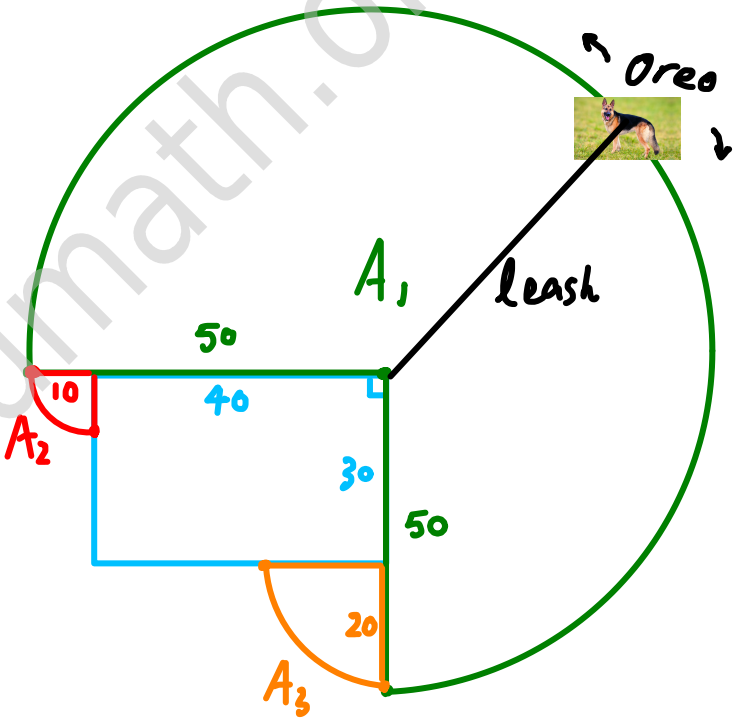
- Cut a circular pizza into even number of equal slices.
- The slices can be rearranged to form a nearly rectangular shape.
- The nearly rectangular shape becomes a rectangle if there are infinitely many slices.

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* Area problems involving circles

- My rectangular house is 30 feet by 40 feet. My dog Oreo is outside on a 50 feet leash that is tied to a corner of my house. What is the area of the region in which Oreo can roam?

Note: Draw a good diagram to illustrate the scene.



[Soln] The region consists of 3 parts with areas:

$$A_1 = \frac{3}{4}\pi(50)^2$$

$$A_2 = \frac{1}{4}\pi(10)^2$$

$$A_3 = \frac{1}{4}\pi(20)^2$$

Total area:

$$A = \frac{3}{4}\pi(50)^2 + \frac{1}{4}\pi(10)^2 + \frac{1}{4}\pi(20)^2$$

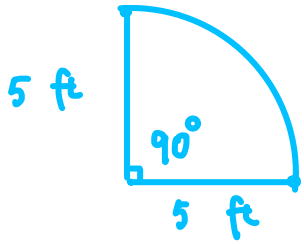
$$= \frac{1}{4}\pi \cdot 10^2 \left(3 \cdot 5^2 + 1 \cdot 1^2 + 1 \cdot 2^2\right)$$

$$= \frac{1}{4}\pi \cdot 100\,(75 + 1 + 4) = 25 \times 80\pi$$

$$= 2000\pi \ [\text{ft}^2]$$

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Practice

7. 

I have a garden with the shape as a quarter circle of radius 5 feet. I need 50 seeds to plant the garden. How many seeds do I need if the radius is doubled? _____

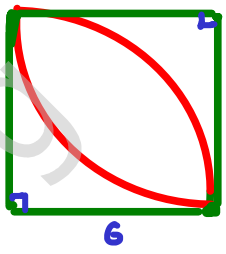
8. My rectangular house is 30 feet by 40 feet. My dog Oreo is outside on a 20 feet leash that is tied to a corner of my house. What is the area of the region in which Oreo can roam? _____ $\cdot \pi$ [ft^2].

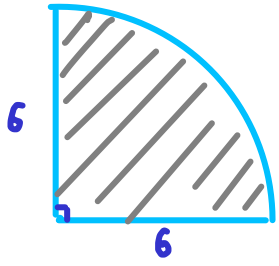
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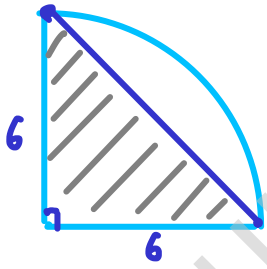
- Two quarter-circles are drawn with their centers at opposite vertices of a square, as shown in the diagram. If the side length of the square is 6, what is area of the region enclosed by the two quarter-circles?

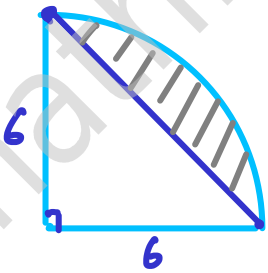
[Soln 1]

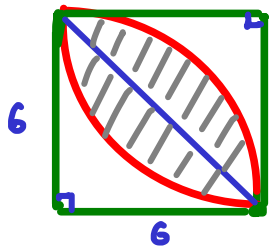
Strategy: divide to conquer











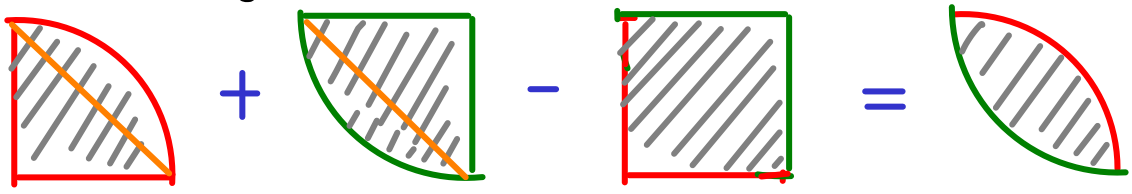
area: $\frac{1}{4}\pi \cdot 6^2$ $\frac{1}{2} \cdot 6^2$ $\frac{1}{4}\pi 6^2 - \frac{1}{2} \cdot 6^2$ $2\left(\frac{1}{4}\pi 6^2 - \frac{1}{2} \cdot 6^2\right)$

Answer: $A = 2\left(\frac{1}{4}\pi 6^2 - \frac{1}{2} \cdot 6^2\right) = 2 \cdot (9\pi - 18)$

$= 18\pi - 36$

[Soln 2]

Strategy: Venn diagram

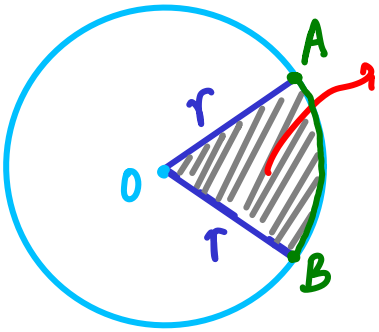


$\frac{1}{4}\pi \cdot 6^2$ $\frac{1}{4}\pi \cdot 6^2$ 6^2

Answer $A = \frac{1}{4}\pi \cdot 6^2 + \frac{1}{4}\pi \cdot 6^2 - 6^2 = 18\pi - 36$

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Arc Length and Area of a Sector



a sector: bounded by two radii and their intercepted arc

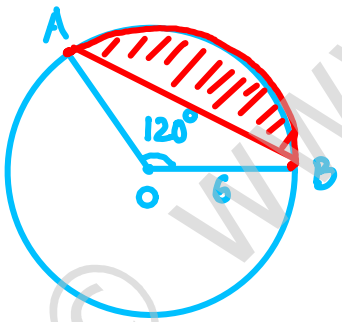
$$\text{Arc length of } \overset{\frown}{AB} = \frac{x}{360} \cdot 2\pi r$$

fraction — circle circumference

$$\text{Area of a sector: } A = \frac{x}{360} \cdot \pi r^2$$

fraction — circle area

E.g.



The area of the shaded region:

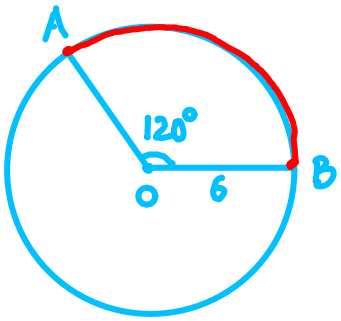
(sector area) − (triangle area)

$$= \frac{120}{360}\pi \cdot 6^2 - \frac{1}{2} \cdot 6\sqrt{3} \cdot 3$$

$$= 12\pi - 9\sqrt{3}$$

Practice

9.



The arc length of $\overset{\frown}{AB}$:

( ) · π

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